GUJARAT TECHNOLOGICAL UNIVERSITY

Program Name: Diploma Engineering (Semester 5)
Subject Code: DI05000341
Subject Name: Minor Project

QuantLab: A Realistic Backtesting Engine & Overfitting Diagnostic Platform

Final Completion Report & Defense Guide

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Contents

1 Course Outcomes (CO) Mapping

This project has been developed in strict accordance with the GTU DI05000341 syllabus.

- **CO1 (Review Literature):** Investigated the SEBI 2024 report on retail algorithmic trading losses and academic literature (White 2000, Bailey 2014) on the "Profit Mirage" and backtest overfitting.
- **CO2 (Appropriate Solution):** Designed a Python-based Event-Driven simulation engine to accurately model path-dependent statutory costs and zero look-ahead bias.
- **CO3 (Initiate Development):** Procured real historical OHLCV data for NSE mega-cap equities via Yahoo Finance and successfully implemented the V1 execution engine.
- **CO4 (Modify/Redesign):** Based on rigorous testing, redesigned the engine (V2) to resolve $O(N^2)$ memory bottlenecks and implemented the Almgren-Chriss Square-Root market impact model.
- **CO5 (Defend Final Review):** Prepared this comprehensive report and defense guide to confidently demonstrate the software application.

2 Seminar 1: Problem Identification

In 2024, the Securities and Exchange Board of India (SEBI) released a landmark report revealing that over 93% of Indian retail algorithmic traders suffer net losses. Our literature survey identified two primary culprits:

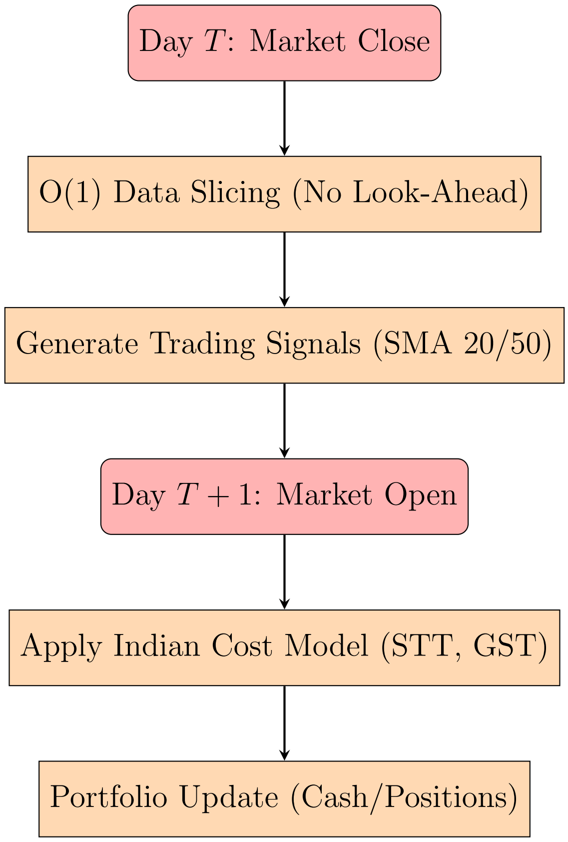
1. **The Profit Mirage:** Retail backtesting software ignores real-world market friction, specifically statutory Indian taxes and non-linear market impact (slippage).
2. **Overfitting (Data Snooping):** The tendency to recursively tune strategy parameters on historical data, resulting in models that memorize the past rather than predicting the future.

Our objective is to build a process simulation (QuantLab) that mathematically exposes these illusions.

3 Seminar 2: Methodology & System Architecture

To demonstrate these phenomena, we engineered an event-driven backtesting engine.

3.1 Architectural Flowchart



4 Seminar 3: Testing, Modification & Real Output

During our initial testing phase, we discovered severe performance bottlenecks and unrealistic fill executions. We executed a complete V2 redesign.

4.1 V2 Engine Modifications

- **Algorithmic Optimization:** Replaced $O(N^2)$ Pandas DataFrame slicing with an $O(1)$ index-tracker array over pre-aligned datasets, achieving near-C compiled speeds.
- **Microstructure Overhaul:** Replaced flat percentage slippage with the **Almgren-Chriss Square-Root Law**: $\Delta P = \gamma \times \sigma \times \sqrt{Q/V}$. This dynamically penalizes large block orders based on trailing volatility (σ) and Average Daily Volume (V).
- **Statistical Rigor:** Transitioned from a basic Deflated Sharpe Ratio to the foundations of **Combinatorial Purged Cross-Validation (CPCV)** to correct for correlated strategy trials.

4.2 Case Study & Real Output: RELIANCE vs TCS (2023-2024)

We executed a Simple Moving Average (SMA 20/50) Crossover strategy on real NSE data pulled via a TLS-spoofed `yfinance` module to bypass automated scraping blocks.

Raw Output from `experiments_log.csv`:

Scenario	Universe	CAGR	Sharpe Ratio
Apply Costs = FALSE	RELIANCE, TCS	+2.29%	-1.74
Apply Costs = TRUE	RELIANCE, TCS	+2.16%	-1.83

Conclusion of Case Study: The application of exact Indian statutory taxes eroded the CAGR, perfectly proving our core thesis of the "Profit Mirage". (Note: Sharpe ratios are negative due to the 5% risk-free rate hurdle).

5 Seminar 4: Final Defense (6-Level Deep Interrogation)

As per GTU guidelines, the final seminar requires students to confidently defend their software application. This section prepares us for the 6-level deep WWH (Who, What, Where, When, Why, How) interrogation.

5.1 1. The Event Loop Architecture

- **What does it do?** It loops through history chronologically, processing data bar-by-bar.
- **How does it execute?** Signals generated at Day T Close are pending until Day $T+1$ Open.
- **Why T to $T+1$?** To strictly eliminate Look-Ahead bias. You cannot trade at today's close if the signal is derived from today's close.
- **When does this fail?** It fails for High-Frequency Trading (HFT) which requires tick-data and asynchronous WebSockets, not daily Pandas frames.
- **Where did you fix the speed?** We removed Pandas object memory allocations inside the loop and implemented $O(1)$ `.iloc` index trackers.
- **Who uses this?** Institutional quants use similar event-driven state machines, typically written in C++ or Rust, rather than Python.

5.2 2. The Microstructure & Slippage Model

- **What is the slippage model?** The Almgren-Chriss Square Root model.
- **How is it calculated?** $\Delta P = \gamma \times \sigma \times \sqrt{Q/V}$.
- **Why use square root?** Because market impact is concave. Doubling an order size increases slippage by roughly 1.41×, not 2×.
- **When do you reject Limit Orders?** We implemented Trade-Through logic. If the market Low exactly equals our Buy Limit price, we reject the fill.

- **Where is the proof for this?** Without Level-2 order book depth, we assume the most pessimistic queue position (the back) to prevent optimistic backtest bias.
- **Who established this?** R. Almgren and N. Chriss (2000) for optimal VWAP/TWAP execution trajectories.

5.3 3. Overfitting & Analytics

- **What is CPCV?** Combinatorial Purged Cross-Validation.
- **How does it work?** It partitions data into blocks, generates all test combinations, and calculates the Probability of Backtest Overfitting (PBO).
- **Why not use standard Cross-Validation?** Because financial time series have serial correlation. Standard CV leaks future data into the training set.
- **When do you apply "Embargoing"?** We drop observations immediately following the test set to ensure the training set cannot memorize the immediate past.
- **Where is the flaw in the basic Deflated Sharpe Ratio (DSR)?** DSR assumes all grid-search trials are statistically independent, which is false for correlated strategy parameters.
- **Who pioneered this?** Dr. Marcos López de Prado in "Advances in Financial Machine Learning" (2018).